

Data Science Group Project Handbook 2025-26

SEMTM0044

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Introduction

The dissertation is an extended report of an independent study that identifies and investigates a particular question and explores it systematically over a sustained period of time. The dissertation provides you with the opportunity to read extensively, complement your learning from the units and apply your data science knowledge and skills to real-world data, to solve problems with potential real-world impact. To do so, you need to work as a group to independently plan, run, document, and present a substantial piece of original work. The aim of this unit is to give you an opportunity to integrate material from all taught units in data science, to demonstrate the breadth and depth of your learning on the MSc, and to prepare you for the type of data problems you may encounter in your jobs after graduation.

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Timeline

Date	Activity
Week of 1 June 2026	Projects Start with Kick-off meetings
Week of 29 June 2026	Formative presentations
4th September 2026	Final Dissertation Submission deadline
Starting from Week of 24 August 2026 (2 week window around 4 Sep). Actual dates arranged with your supervisor.	Final group presentations

Tasks which count towards your unit mark (summative):

Dissertation portfolio), via Blackboard, 1pm Friday 4th September 2026, consisting of:

- A 30-page report (60% assessing all learning outcomes).
- A 2 page Individual reflective account (20% assessing learning outcomes 1,2 and 5).
- A 15-minute group presentation (live online over Teams) about your project(20% assessing learning outcomes 2 and 5).

Dissertation project preparation and support

Your attendance during the dissertation period

Students are expected to be in Bristol for the entirety of the dissertation period. It is essential to regularly attend group meetings and engage in group work. Furthermore, being in the UK is a legal requirement for students who are in the UK on a student visa. If for any reason you need to be away from the UK during the dissertation period, it is essential that you contact the PGT student admin team (semt-student-enquiries@bristol.ac.uk) as soon as possible for guidance.

Dissertation personnel – what you can expect

Your academic supervision team

Each project will be associated with an academic supervisor and a teaching assistant (TA) who will be responsible for the main part of the supervision during the summer. You will have 1 scheduled meeting every month with your supervisor and 2 scheduled meetings every month with your TA.

It is likely that you have already completed a dissertation before at undergraduate level, under the supervision of a dissertation supervisor. Although there will be similarities between the dissertation supervision you received then and will receive now, it is very important for you to understand that there will also be differences – primarily due to the fact that, at postgraduate level, you are expected to work more independently.

The bullet points below provide guidance on what you can expect from your academic dissertation supervisor. Please note that they are intended as general expectations and are therefore not formally binding.

- Your supervisor can help you with:
 - Narrowing the focus of your research and formulating your research/technical question(s),
 - Aims, objectives and outcomes
 - Discussing research ethics approval and granting approval in many instances
 - Guiding some of your reading
 - Advising on any stage of the research process and any difficulties you may face
 - Assisting in structuring your dissertation
 - Commenting on draft material and advising you on final submission.
- Supervision time:
 - The academic supervisor and the TAs form the supervision team during the summer. .
 - You can expect about 3 dissertation supervision meetings with your supervisor and 5-6 meetings with the TAs..
 - You can expect about one substantial email from your supervisor per month. Your dissertation is a piece of independent postgraduate-level work, and as such, it is important for you to remember that the role of your supervisor is

mainly to provide **guidance**. It is not their role to provide full-time dissertation support.

- Feedback on your dissertation draft: You can expect your supervisor to provide substantial feedback on **one** draft of your dissertation. To receive this feedback in a timely manner, it is extremely important that you:
 - Give your dissertation supervisor at least 14 days to provide the feedback.
 - Agree the date of your submission of the draft with your supervisor in advance, so that it does not coincide with any leave that you or your supervisor may have booked over the summer.

If you do not follow both of these precautions, then you run the serious risk of causing a delay to your feedback.

- Your ethical approval form (information about ethical approval is found further below): You need to speak with your supervisor about your ethical approval form. You can expect your supervisor to guide you with this process. It is your responsibility (*not your supervisor's*) to ensure that your ethical approval form is submitted in time. TAs will monitor your engagement, group contributions as well as provide technical supervision support during the 3 months.

Your industry partner

Each project has an industry partner who acts as additional supervisor. Meetings with the industry supervisors will be arranged separately if needed and may take on different forms. Your industry partner can give feedback and ideas to shape your project in addition to your academic supervisor.

Please do not directly contact the industry partner. If you have any questions about the project, please first contact the TA or your supervisor. If questions need to be asked from the partner, your supervisor will arrange to forward them to the partner.

Supervision Meeting Schedule

Your meeting schedule will be arranged by your supervisory team (academic and TA), and should be communicated to your team on the first two weeks of the project. On each month, you can expect three scheduled meetings: one with the academic supervisor and two with the TA per month. On each month, there is also a possibility of having **one** additional (optional / ad-hoc) meeting with the academic supervisor if there are urgent/critical issues that need to be addressed. Please agree with your academic supervisor on how to your group could flag that this optional meeting may be needed.

All supervisory meetings are intended to be held in-person unless otherwise communicated. Please make sure you are prepared for each meeting. It is essential that you attend all supervisory meetings on the dates agreed with your supervisory team. Please note that your supervisors and TAs might be away for 1 or more of these meetings due to holiday/conference travel and so on, in which case alternative meeting arrangements will be agreed with your teams.

Guidance for your formative presentation (late June)

In late June, you will present your preliminary results to your academic supervisor and other audience members such as other academics, industry partners, or your peers. This presentation is a chance to showcase what you have worked on up until that point and to receive feedback for the final stages of your dissertation work. The presentation will consist of a given time for you to present your work as well as reserved time for questions and discussions at the end.

We encourage all group members to take ownership of one part of the presentation and present it to the audience. However, you are free to decide a different model of division of labour. If you do decide that not all group members are to present a part, you need to clarify in your presentation who contributed what to the presentation.

Guidance for your final group dissertation report

Your final group dissertation report will present your analysis and results and set it into context of the relevant academic literature. You can also create an executive summary that can be shared with the industry partners.

Length of your report and penalties for exceeding length

The maximum length of a dissertation is 30 pages. This includes everything from the start of your first chapter till the end of your last chapter. It excludes front matter, bibliography, references, contents pages, declaration, tables, appendices etc. There is no minimum word count for the dissertation. However, a dissertation that falls substantially short of the page limit is unlikely to have covered the subject in sufficient depth.

Dissertations that exceed the maximum of 30 pages as outlined above will be subject to the following mark deductions:

- 5 marks per page beyond the limit

Due to the heavy penalty on extra pages, please ensure you keep your content within the 30 page limit.

A LaTeX template for the report will be provided and using LaTeX is highly recommended for the report.

Penalties for late work

- Dissertations submitted after the deadline and without a permitted extension will be subject to late penalties in accordance with the [2025/26 Regulations and Code of Practice for Taught Programmes \(PDF, 1,005kB\)](#)

If you encountered difficult circumstances that prevented you from submitting your dissertation on time, you should complete an exceptional circumstances form and provide evidence.

Guidance for your Individual reflective account

This should be no more than two pages of A4, and the text should be no less than 500 words. It is an opportunity for you to reflect on the project experience, the positive and negative aspects of the technical work itself, and of working in a small team to a tight deadline.

It must cover these components:

1. A visualisation of your individual project activity from the Github repository/log data
2. A weekly log of activities/tasks you have done.
3. Your individual reflections of working on this project. This can include, what worked well, what did not work well, what could have improved, what did you learn (both technical + non-technical skills).

Guidance for your final summative presentation (late August)

In late August, you will present your final results to your academic supervisor and other audience members such as other academics, industry partners, or your peers. This presentation is your chance to showcase what you have achieved in your dissertation work. The presentation will consist of a given time (15 min) for you to present your work as well as reserved time for questions and discussions at the end.

For the final presentation, we expect all group members to contribute in the presentation.

Passing EMATM0044

To achieve a pass on EMATM0044 you must achieve an overall unit mark of 50% or more.

Reassessment on EMAT0044

To be eligible for reassessment on EMATM0044, your overall unit mark must be between 45% to 49% inclusive. You will be required to take reassessment for the assessment components that you have failed.

If reassessment is required, then you will prepare an individual written report that will be assigned a weight equal to the sum of the weights of any components that you did not pass (e.g., if you passed the group presentation but did not pass the group written project or the individual reflective account, then the reassessment would be assigned a weight of 80%). The reassessment will involve working with the dataset that was originally given to your group and/or reflecting on the work done throughout the unit. For

your individual reassessment, you may be asked to develop and critique your group's original submission (including highlighting improvements that could be made to the group report) or you may be given a different task associated with the original dataset or project. The precise form of the reassessment will depend on which learning outcomes were not successfully demonstrated in your original assessments.

If you have further questions, you are welcome to discuss these with your Dissertation Supervisor and the Dissertation Team.

Failure on EMAT0044

If your EMATM0044 overall unit mark is below 45% (i.e. 44% or less), you will not be eligible for re-assessment for EMATM0044 Dissertation Project. This would also mean that you would not fulfil the requirements to graduate with a full MSc, and you would be required to withdraw (RTW). If you are required to withdraw, you may still be eligible to receive the award of Postgraduate Diploma (PGDip) or Postgraduate Certificate (PGCert). Some exceptions may be possible if you have registered Exceptional Circumstances with sufficient evidence of the impact on your ability to study.

AI Policy

Q: Can I use AI in my dissertation project?

AI is an important area of technology development that is highly relevant to data science. However, it is important to maintain academic integrity when approaching the use of AI within your dissertation and other work. Undisclosed use of AI is prohibited, meaning you must declare any use of AI in your work. Your supervisor will check your work for academic integrity including undisclosed use of AI by reviewing the supplementary materials submitted as part of your portfolio.

The University has four Artificial intelligence (AI) assessment categories

<https://www.bristol.ac.uk/students/support/academic-advice/using-artificial-intelligence/>:

- Category 1: prohibited (not allowed): You cannot use AI tools at all, including spelling and grammar checkers.
- Category 2: minimal (a little): You can use AI tools for spelling and grammar check. You can use it for rewording occasional phrases, but not sentences, large parts of text or paragraphs. It should not change the meaning or the context of your work. If you have not been given any other instructions, use this as the default category for your assessment and studying.
- Category 3: selective (certain tasks): AI tools can be used in certain ways. Your lecturer will tell you what you can use it for. However, you do not have to use AI tools in your work. You will not be given extra credit for using it.

- Category 4: integral (important): AI tools are needed for the assessment. You will be told which tools to use and in what capacity.

The dissertation portfolio (report + presentation + code/supplementary material) is Category 2. You may use third-party AI tools to support, but not replace, your individual academic contribution. This means:

- You may use third party AI tools for a) spelling and grammar check, and b) rewording/minor translation; but not for generating large sections of text, which is considered a form of cheating.
- The following example uses are inappropriate and will be considered academic misconduct:
 - “Write my literature review on ML techniques used in topic xyz”
 - “Generate the methods section about my DL implementation”
- “Produce my results section” Use of third-party AI tools should not change the meaning or the context of your work as this means that you are not demonstrating your own understanding of the algorithms used, experiments undertaken, or design decisions.

The project work is Category 3. You may use AI as relevant to the technical content of your project.

This means:

- You may use AI to support your research. Examples:
 - o Debugging advice: “Why is my Python script crashing? Here’s the error log.”
 - o Concept clarification “Explain the difference between EKF-SLAM and FastSLAM2.0.”
 - o Generating example code snippets that the student then adapts, for instance, “Give a minimal python example showing use of k-means on the given data”
 - o Researching appropriate techniques: “Suggest appropriate statistical tests for evaluating robot localisation accuracy.”
- Students should consider that only their individual contribution to the dissertation will be used to generate your grade. Just as you disclose work done by individual team members, work done by third party AI must be disclosed and will not be considered part of their individual contribution.
- Third-party AI tools should not be used to fabricate experimental results/data.

Prohibited Uses!!!

You are not permitted to use AI tools for any other aspect of the dissertation. In particular, AI tools must **not** be used to:

- Generate or rewrite substantive content (e.g., research questions, literature review, theoretical discussion, argumentation, interpretation of results, or conclusions);
- Design the overall structure or line of argument of your dissertation;

- Draft, edit, or rewrite text beyond the limited spelling and grammar checking permitted above;
- Generate citations or reference lists.
- Using AI for purpose beyond permitted uses may constitute academic misconduct under University regulations and may result in serious penalties.

How should I disclose use of AI?

Students must include a statement in the frontmatter of their dissertation:

- Third-party AI was not used in this project.
- Third-party AI was used in the following ways within this project... [state use of AI]

Reliability and Responsibility

AI output can be incomplete, biased, or incorrect and must always be treated with caution.

- You should independently verify any information, suggestions, or code obtained from AI tools.
- You must understand and be able to explain every part of your dissertation text and code.
- **You should not submit anything you are not prepared to take full responsibility for. You are not required to use AI tools, and no additional credit is awarded for doing so.**

Suggested plan for your final group extended dissertation report

Organisation of chapters

It is usual for the dissertation to be structured into chapters, each fulfilling a particular role in building your argument, demonstrating your understanding of the issue, your understanding of others' related research and exploring your findings. The chapters need to be carefully structured with an introduction to help steer the reader, and a summary to remind the reader of the key themes, concepts, ideas or issues that were covered and to be taken forward. The chapters should be organised and sub-divided to support the reader's ability to follow your arguments as they are introduced and developed. However, it is important not to over-organise a chapter to include so many headings and sub-headings that the flow of the text is fractured.

The abstract

A succinct and clear outline setting out the problem investigated, how it was investigated, the major findings and conclusions.

Introduction

The introduction should briefly outline the purpose of the study and offer the reader an introduction to the substantive area. It should discuss the rationale for the study,

providing a clear statement of the research questions, aims, objectives and outcomes, explaining why it was selected, and why it is relevant and important to you. The introduction should forewarn the reader what he/she will be reading. It should therefore outline briefly what the dissertation is about and the ways in which the research was undertaken and any key issues or concerns presented by the dissertation. An early rough draft, which you share with your supervisor, can be a useful way of clarifying your thoughts and intentions and keeping you focused as your study progresses.

Research philosophy and methods: A description of the research study

This chapter describes what informed your research design and why and justifies the choices you have made. It should describe the selection and subsequent development, where appropriate, of your research tools and the role that piloting may have played. It should be reflective, in that it might comment on limitations of the research as it was finally executed, or on any differences between your original plans and what subsequently proved possible.

Literature review

The purpose of the literature review is to present for the reader a concise overview of literature relevant to your research questions, aims, objectives and outcomes. You should set out the key research themes and main arguments (relevant to your research questions). A description of what others have said is not enough. You should critically evaluate what they have said, maybe by asking yourself whether their research was done in one place, in a particular way, with particular subjects, and is relevant to a particular context. The coherent argument in your literature review will be demonstrated in the way you use the work of others to underpin your research questions, aims, objectives and outcomes and research methods.

The literature review also demonstrates a number of skills to its readers. It should show accuracy of reading and of reporting. It should show accuracy in citing references and in quoting material. It should demonstrate skills in using and commenting on cited materials in developing an argument. The chapter should demonstrate the ability to synthesise a volume of material and to sift out key themes. The chapter should demonstrate an ability to evaluate differing sources, recognising the limitations of using sources such as Wikipedia and Google Scholar versus ISI rated materials and high-quality academic journals identified and sourced through credible research data bases such as Web of Knowledge and Web of Science. It should show ability to write a balanced review and not simply to highlight sources that support a favoured perspective.

Presentation of findings and their analysis (may be in separate 'findings/results' and 'discussion' chapters)

This chapter, or chapters, allows you to describe the responses to your research instruments and then to analyse the significance of what you found. The chapter(s) will need careful structuring. You will have to decide how you are going to organise your findings, to present them most clearly and succinctly to your readers. You will need to take into account, for example, whether to work by theme, by constituency, by organisation or by geographical area. Whatever the organising principles, these need to be explained to the reader and the rationale for particular choices made clear.

Although you must describe your data clearly for your readers, it also needs to be presented and analysed. This means comparing constituencies, exploring similarities

and differences and trying to account for these. If you use interview material, then you must use quotations to illustrate carefully made and criticised points. You should not simply present your reader with great chunks of quotes assuming that they will make a case for you. It means offering appropriate statistical analysis of the data where appropriate and moving beyond a simple rehearsal of percentages and diagrammatic presentations of data. At this point (and perhaps in a separate chapter) you should compare your data against that offered by other researchers cited in your literature review, seeing where it supports or moves away from others' work, and try to account for this. The strength of the links between this chapter and the literature review enhances the dissertation.

It is important to revisit your research questions, aims, objectives and outcomes to assess the extent to which you have achieved your expectations, fully justifying and explaining your conclusions.

Summary and conclusions

This is the place to pull together the key themes of the dissertation and draw out implications for practice or recommendations for further research in the area. It is not the place to introduce new themes. A reader should be able to read the conclusion and have a clear sense of what the dissertation has been about, what has been discussed within it, and what you have made of it.

References

This is a list of all the books, articles, or web sites you have referred to as citations in the dissertation. This is not a bibliography. Quotations should be copied correctly. You are advised to make a note of all references as you read them rather than trying to go back through everything you have read at a later stage.

You should reference using a consistent referencing system, such as the IEEE referencing system; you should normally cite sources in the text. As a general rule, you should avoid using footnotes to reference.

If you include a quote, it should be in quotation marks, and a page number included in the in-text reference. Whilst you should normally avoid larger quotes, if you include them, you should also indent the text. If you cite a paper in your essay, you should also include a full reference to the paper in the reference list at the end of the paper.

Do not list papers in your reference list that you have not referenced in the paper.

Appendices

Your supervisor will advise you about what material you should put in appendices. Normally, this is where you present sample research tools, examples of data, perhaps examples of coding and analysis, sample teaching materials etc., but each study will have its own appropriate selection. You do not normally need to include all of your original data in your appendices, but you must make sure your original data is available to be sent to examiners, if, and only if, requested.

Academic integrity

Turnitin will be used by markers to determine the similarity index of your submission document. The similarity index is a primary indicator but does not directly indicate plagiarism. Unacceptably high similarity indices prompt further detailed investigation focusing upon the provenance of your material and your adherence to accepted

conventions for presenting written academic material, including your use of citations and references.

What if I need to apply for an extension?

Extensions must be applied for through our admin team (School of Engineering Mathematics & Technology Student Mailbox semt-student-enquiries@bristol.ac.uk) and are only granted for exceptional circumstances (ECs). To submit ECs, the guidance and a link to the form can be found here:
<https://www.bristol.ac.uk/students/support/academic-advice/assessment-support/exceptional-circumstances/>

You will need to submit evidence. Losing data due to a computer failure or loss does not count as an EC. Therefore, make sure you back up regularly your project in multiple ways (e.g., OneDrive, Dropbox, GitHub, manually on an external hard drive, etc.). If using GitHub, please ensure your repository is set to “private” during the period of your dissertation project – if someone accesses and used work that you have shared publicly, this may result in a plagiarism offense registered against you. Students are automatically given an extension to their dissertation if they have reassessments. The length of the extension depends on the number of credits you are resitting, see <https://www.bristol.ac.uk/academic-quality/assessment/regulations-and-code-of-practice-for-taught-programmes/extension-pgt/>

Cloud Resources

We can provide full AWS accounts (not student limited accounts) with \$100 credit to each team if needed. To obtain these, discuss with your supervisor if accounts are needed. If your supervisor agrees then use the form provided to apply for an account. One account will be created per team and one person will initially get full access, they can then add the other members of the team on this account.

Form: [Form requesting AWS cloud GPU resources 25-26 \(Group Projects\) – Fill in form](#)